Python for computational Problem Solving · Unit_02 · Unit 2: Collections & Functions

DRAFT — UE26CS151A · Unit_02 · Unit 2: Collections & Functions · Dictionary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Dictionary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Dictionary

1 Dictionary

Study focus: Dictionary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Let me repeat, a set is an unordered collection of unique elements with zero or more elements present in it.

Key details

- I'll say S is equal to this.
- open curly breeze and closed curly breeze.
- And I want to find out what is the type of yes.
- And if I say what is a type of S, but basically say it as a dictionary.
- So which means my dictionary as well as set, we've in order to create them, we are going to be making use of curly braces itself.
- I need to work out what the set.
- Find out what is the type of S.
- But if your intention is to create an empty set, then make use of your constructor called as set in order to represent an empty set.
- Go to be set.
- He is called as a constructor.
- You can use constructor in order to create an empty set.
- Has got these some attributes.
- I'll be using a conventions yes, s1, s2, s3 in order to represent it as a set.
- The first attribute is that set elements are unique in nature.
- Set elements are unique in nature.
- What does it mean?
- I'll say two, one, four again, and a three again.
- And if you can see I've got one, two and three and four.

- Okay, so which means elements in a set are unique.
- The elements are hashable in nature.
- Elements are hashable in nature.
- That's all we need to know at this particular point.
- Fine, if it has an hash value which never changes during its lifetime.

Example or application from the lecture

- So let's create our first set.
- At this point, for our level, we just need to understand that hashing is a mechanism used to convert a given element into an integer.
- Hashing is a mechanism used to convert a given element into an integer.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

Fine, which also means that it needs to have a function called as underscore hash underscore as one of its method.

Key details

- Data structure having this particular method, then we call it as what it is hashable in nature.
- And if you can see and hashable method if it is available then we basically call it as what?
- It is uses a mechanism called as hash method.
- It is clear to you it's a mechanism that is used in order to.
- The ones that are not hashable are list, dictionaries, and all of them.
- Sorry I'm creating a set.
- I'll say set.
- I'll say L of S.
- This is five.
- What is the minimum element in this S?
- Minimum element is one.
- And I can I do max of yes.
- Max is element is five.
- I also see some of yes.
- And I'll do one more thing.
- I'll create something like this.
- Yes one is equal to hundreds twenty two.
- When I say S1, if you can see, if you can see, it's evident that set elements are unordered in nature.
- The order of the set cannot be determined.
- An important attribute is that the set elements are unordered in nature.
- We can never assume the order of elements.

- The reason is the way in which I have entered and the way in which it is present internally is different.
- because of which because of which I cannot do something like this.
- I cannot do because it does not support indexing.
- The main reason is the set elements are unordered in nature.
- I'll see S1.
- dot add and I'll say two hundred.
- This is fine.
- I said s one dot add which is two hundred.
- You could see S1 where 200 is being added.
- The moment you say add, it adds a well.
- Where does it add a value?
- We never know because it is based on the concept of hashing, which means sets are mutable in nature.
- sets are mutable.
- I can change the contents after it has been created.
- Remember, please always remember.
- Set is mutable, but it elements are.
- Its elements are immutable.
- Set is mutable in nature, but its elements are immutable.
- It means that let me create this.

Example or application from the lecture

- Convert a given element into an integer.
- Let's go ahead with the next one.
- Or before that let's find out what are the common operations that we could do it on this set.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

One, two, three, and I'll create a list within this.

Key details

- So let me do this and it says unhashable type list, which means my set has a whole is mutable but its elements are immutable, which always requires you to have the values that are hashable objects.
- Set is mutable, it is unordered, duplications are not allowed, it is unique in nature, okay.
- It does not support indexing.
- As it does not support indexing, I cannot also perform slicing operation.
- This is my S L C for I N S print I.
- And see I've got the values, which means set is iterable.
- Set is iterable.
- What is the use of a set?

- Sets are basically used in order to remove the duplicate elements, which in turn allows us to have or find the unique elements that are present.
- Sets are very useful in order to remove the duplicate elements.
- The moment I remove the duplicate elements, sorry, duplicate elements, I'm going to be having the unique elements that are present.
- I can also compare.
- Two I trables for common elements or common differences.
- Is this clear?
- We will also be working on this particular concept.
- And there's one more thing that I want to.
- I'm talking about a membership operator.
- Returns true if the element is present in set.
- returns false if the required element is not present in a set.
- Okay and I can also do not in it's true because it is not present in the set.
- I'll say D I R of sets.
- And you can see these are the functionalities that are available.
- I'm going to be showing you a few of them.
- The rest of them are left for you to understand.
- In order to do, I will try to make use of two different sets, and the values of these sets are.
- Five and six.
- And I will have my S two being.
- Six, seven, eight.
- Want some common elements also, so I will retain five because I can have it in any order, so order doesn't matter.
- I'm trying to perform concatenation operation.
- Set and set concatenation operation is not supported.
- Can I do this?
- S1 plus two.
- Says still concatenation operation is not supported.

Example or application from the lecture

- Let's find out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

Is it fine?

Key details

- I'll say S1 [ASR low confidence; check audio] dot union of estop.
- I'll say s1 dot union of s two, which means I am combining the values of s one and s two.
- Returns the union of sets as a new set for us.

- I can also say this as one, you can also make use of this particular symbol in order to represent it as what?
- The union operation.
- I can also do something like this.
- S1 minus S2, the ones that are present in S1 and the ones that are present in S2 have been removed from S1.
- So according to this one, two, three and four because five six is here, five six is also here.
- The other way is also true for this.
- I can also say S1 minus S2.
- Remove the ones that are.
- There in both S1 and S two and give the elements of S two only.
- And if you can see even though we said seven, eight, nine and ten, the order has not been maintained.
- [ASR low confidence; check audio].
- This is my S1.
- We will see S1.Remove.
- S1 dot remove.
- Which one shall we remove?
- We'll remove three from here.
- Is this fine?
- What does a remove operation will do?
- It removes the specified item in a set.
- If an item is not present, okay, now we have already removed three.
- I'm trying to remove three again.
- It throws us an error.
- Otherwise it throws us an error.
- Also do discard.
- Discard four is what we will do.
- We'll try to find out.
- What does this discard will do?
- I'll say yes one.
- It has also removed.
- Remo also did the same thing for us.
- I will try to remove the same item again.
- And in this case, it does not throw me an error.
- Removes a particular element.
- Discord also does the same thing.
- If an item is not present, remote throws us an error, but whereas discard does not throw us an error.
- And S1.pop is an operation.
- It pops out some element.
- In this case it has popped out one.
- Let me pop out one more.
- It has popped out five, in this case it has popped out six, and so on.

- Removes an element or an item in a set.
- In this case, we are lucky enough that the pop operation is removing elements one by one.
- Is it guaranteed?
- No, it is not guaranteed.
- The reason is sets are unordered.

Example or application from the lecture

- Let's find out what does it do.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 20:00–24:59

There are a few more things that you can always work out with the set.

Key details

- I've just shown you a few of them.
- I request you all to try out be the other ones because they're all self explanatory.
- Is this fine?
- Symmetric difference, symmetric difference update.
- Wherever you have something called as an update.
- It always does the changes to an existing set.
- Please remember, whenever you have an update, operation is performed at the same time where it is being done, it is going to be done to that particular set itself.
- Is this fine?
- I request you all to try out with the other functionalities that are present.
- keep performing the operations.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Dictionary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_02/AV_Summaries/034_Dictionary__9c4cb33a65041296.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Dictionary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_02/AV_Summaries/034_Dictionary__9c4cb33a65041296.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Dictionary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_02/AV_Summaries/034_Dictionary__9c4cb33a65041296.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9998; only the JSON segmentation supplies the displayed timestamps.
Dictionary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CS151A_Python_for_computational_Problem_Solving/Unit_02/AV_Summaries/034_Dictionary__9c4cb33a65041296.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 3 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00007	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00009	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

C Revision and retrieval practice